

# MATTHIAS MASIERO

908-432-5309 | [matthiasmasiero0@gmail.com](mailto:matthiasmasiero0@gmail.com) | [linkedin.com/in/matthiasmasiero](https://linkedin.com/in/matthiasmasiero) | [github.com/MatthiasMasiero](https://github.com/MatthiasMasiero) | [matthiasmasiero.com](https://matthiasmasiero.com)

## EDUCATION

### Santa Clara University

*Bachelor of Science in Computer Science, Minor in Physics*

Expected June 2028

*Santa Clara, CA*

## EXPERIENCE

### Machine Learning Researcher

*Santa Clara University (Dr. Smita Ghosh)*

Nov 2025 – Present

*Santa Clara, CA*

- Benchmarked transformer classifiers (BERT, RoBERTa, MentalRoBERTa, ELECTRA, clinical BERT variants) across 8 public mental health datasets for early self-harm and crisis-signal detection; models at 0.94–0.98 F1 in-domain collapsed below 0.40 F1 on a curated adversarial test set, exposing reliance on surface lexical cues.
- Leading the research team building transformer baselines for conversation-level risk detection on a synthetic multi-turn dialogue corpus engineered to remove the lexical shortcuts identified in prior benchmarks.
- Contributing models and evaluation to a paper on context-aware safeguards for conversational AI, accepted to ASONAM 2026 (Springer LNCS).

### Sports Data Scientist

*Santa Clara University Athletics*

Jan 2025 – Present

*Santa Clara, CA*

- Processing 750K+ biometric data points per session across 22 athletes at 10Hz sampling frequency from Catapult GPS/accelerometer wearables to predict and prevent non-contact injuries.
- Engineering feature-extraction pipelines and applying regression and classification algorithms to optimize training loads, targeting a 70% reduction in non-contact injuries.
- Built an athlete workload and injury-risk dashboard currently used by SCU teams and training staff; leading its adaptation and rollout to additional teams.

## PROJECTS

### DevAngel | AWS Lambda, Step Functions, Bedrock, CloudWatch | 2025 AWS Hackathon Winner

- Built a real-time incident analysis platform using AWS CloudWatch Logs, Lambda, Step Functions, API Gateway, Bedrock, and EC2 to aggregate metrics and detect and respond to anomalies across production systems.
- Integrated an LLM-powered AI agent (Amazon Q for GitHub) that autonomously generates code-level fixes and opens pull requests when incident root causes are detected.

### Low-Latency Order Matching Engine | C++, kqueue/epoll, TCP

- Engineered high-performance order matching engine in C++ achieving 300K orders/sec with 3.3μs mean latency; profiled and optimized hot path for 30x speedup using cache-aligned structures, SPSC ring buffers, and zero-allocation design.
- Built production-ready event-driven TCP server (kqueue/epoll) with non-blocking I/O and a binary protocol; load tested with 3M+ orders and zero data loss.

### CardTempo | Next.js, TypeScript, Supabase, Stripe, Redis | 2nd Place, SCU Winter Challenge

- Built a profitable fintech platform improving credit scores by 15–160 points; full-stack Next.js/TypeScript (25K+ LOC) with Stripe subscriptions, Supabase PostgreSQL with row-level security, and Redis rate limiting.
- Engineered card recommendation engine using fuzzy string matching, multi-criteria optimization (6 weighted factors), and spending cap modeling across 11 categories with signup bonus attainability checks.

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, TypeScript, JavaScript, SQL, Swift, OCaml, Qiskit, OpenQASM, HTML/CSS

**Tools & Technologies:** AWS (Lambda, EC2, S3, SageMaker), Docker, CI/CD, PyTorch, TensorFlow, Git, Supabase, Stripe, LangChain, Claude Code

## LEADERSHIP

### ACM, Santa Clara University

*Event Coordinator*

Oct 2024 – June 2026

*Santa Clara, CA*

- Board member of the largest CS club on campus; led AI and quantum computing workshops and planned two annual hackathons with 350+ participants and multiple guest speakers.

### Zeta Beta Tau Fraternity

*Treasurer*

2025 – Present

*Santa Clara, CA*

- Manage a \$150K annual budget for a 100-member chapter; automated dues collection and financial reporting.